

Key Points:

- The discrete and empirical troposphere delay models of the VMF data server is comprehensively evaluated using the global International GNSS Service (IGS) data
- The Site-wise of discrete model exhibits the best accuracy and is recommended first, followed by the Grid-wise and the empirical model
- Each model shows different influences by the latitude, elevation, seasons and epochs

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Comparing Discrete and Empirical Troposphere Delay Models: A Global IGS-Based Evaluation

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Abstract Zenith tropospheric delay (ZTD) is an important atmospheric parameter in radio-space-geodetic techniques such as Global Navigation Satellite System (GNSS), which is pivotal for GNSS positioning, navigation and meteorology. The Vienna Mapping Function (VMF) data server is a widely utilized source for implementing ZTD, offering two types of models, that is, the empirical one and the discrete one with Grid-wise and Site-wise models. Therefore, to evaluate the accuracy of these models becomes the focus of this article. Specifically, this study investigates their performances in terms of calculation of ZTD, using the hourly values derived from the International GNSS Service data as references. The results show that the root mean square error (RMSE) of the Site-wise, Grid-wise and global pressure and temperature 3 model are 11.71/13.03/38.56 mm, respectively, indicating the discrete model performs generally better than the empirical model, and the Site-wise model is the better of the two discrete models. From the perspective of spatial resolution, the performance of these three models in ZTD calculation shows obvious influences of latitude changes and elevation differences. From the temporal analysis, the accuracy of the discrete model shows differences over different UTC epochs, while the empirical model can only express the seasonal ZTD characteristics with the average RMSE at different epochs being similar, the specifically values are 39.67, 39.26, 39.38 and 39.18 mm at UTC 0:00, 6:00, 12:00 and 18:00, respectively. The histogram and boxplot well indicate the accuracy differences of the three models in different seasons. Additionally, the time series of three models at different latitudes were also explored in this research. These explorations are conducive to the selection of appropriate models for calculating ZTD based on specific requirements.

1. Introduction

Due to the interaction with water vapor particles and dry gases, the electromagnetic wave signals are subject to refraction and bending when propagating through the neutral atmosphere from the satellite to the receiver, which resulting in a delay known as the troposphere delay (Bevis et al., 1992; Huang et al., 2023). Compared to vacuum propagation, the magnitude of troposphere delay can range from 2 to 20 m depending on the elevation angle of the satellite (Yang, Meng, et al., 2021). Therefore, troposphere delay is a significant source of error in various space geodetic techniques such as Global Navigation Satellite System (GNSS), Very Long Baseline Interferometry (VLBI), and satellite altimetry (Shi, Ouyuan, et al., 2023; Zhao, Su, et al., 2023).

Zenith Troposphere Delay (ZTD) is a commonly used parameter to describe the effect of the troposphere on signal propagation, which is obtained by projecting the slant delay on the propagation path of GNSS signals onto the zenith direction through a mapping function. In GNSS data processing, the use of accurate ZTD values and appropriate constraints can improve convergence time and accuracy (Shi, Li, et al., 2023). In addition, ZTD plays a significant role in climate research and weather forecasting, and can be included in numerical weather prediction (NWP) models. For instance, ZTD observations derived from ground-based GPS receivers have been assimilated operationally into the Met Office North Atlantic and European (NAE) NWP models by the UK Meteorological Office (Bennitt & Jupp, 2012), and the implementation of the ZTD assimilation was also conducted by the French Meteorological Office in a global Four-Dimensional Variational (4DVAR) assimilation and forecasting system (Poli et al., 2007). On the other hand, the accurate and direct measurement of ZTD is still difficult for these real-time or near real-time applications, developing highly accurate, stable, and reliable ZTD models is a crucial focus in the field of GNSS research, and is essential for other space geodetic techniques.

Two types of ZTD models are commonly used in the field of GNSS research (Huang et al., 2021): (a) The meteorological parameter models, such as the Hopfield model (Hopfield, 1969), the Saastamoinen model

(Saastamoinen, 2013), and the Black model (Black & Eisner, 1984). These models rely on measured meteorological parameters of the stations and can achieve an accuracy up to the centimeter level (Zhao, Wang, et al., 2023). However, most GNSS stations are not equipped with atmosphere sensors or have no nearly available meteorological stations, making it impractical to use this type of model directly to obtain ZTD values. (b) The non-meteorological parameter model, which estimates ZTD based only on the location and time of the site. Collins and Langley proposed the UNB model, which has been continuously modified and developed over time, for example, the UNB model, including UNB3m (Leandro et al., 2008), UNBw.na (Leandro et al., 2009), and EGNOS (Ueno et al., 2001). In addition, the IGGtrop series models (W. Li et al., 2018) and the GZTD series models (Yao et al., 2016) were successively proposed based on the analysis of the long-term ZTD values from the meteorological reanalysis data. For this type of model, Yang et al. (2021) have conducted an accuracy assessment of nine traditional ZTD models including UNB3 and EGNOS, Yuan et al. (2024) selected the three representative ZTD models (the global pressure and temperature 3 (GPT3) model, the SHAtropE model and the IGGtropSH model) to conduct the comparison using the data from Crustal Movement Observation Network of China (CMONOC) as references.

The Vienna Mapping Functions (VMF) data server maintained by the Department of Geodesy and Geo-information, TU Wien, established two types of ZTD models, namely the discrete tropospheric delay models and the empirical tropospheric delay models, which have been welcomed by many scholars for their demonstrated good accuracy and versatility. For the discrete models, the ZTD parameters are determined for every six epochs (UTC 00:00, 06:00, 12:00, 18:00) and need to be interpolated for the other epochs (Boehm et al., 2006). Based on the empirical grid coefficients, the empirical tropospheric delay model, namely GPT3 model, provides ZTD using the trigonometric functions (Boehm et al., 2007). So far, the ZTD values derived both from the two types of the tropospheric delay models have been applied in various practice by many scholars such as the development of ZTD models, different GNSS applications and accuracy assessment: (a) For ZTD model development, Liu and Yu (2017) proposed an algorithm for global troposphere delay determination by the combination of GPT/UNB3m. F. Li et al. (2023) developed a zenith wet delay (ZWD) calibration model based on VMF3 to obtain more stable ZWD values globally. J. Li et al. (2023) established an improved ZTD model in Mainland China based on the GPT3 model. Zhang et al. (2023) used global tropospheric delay grid products of VMF1 and VMF3 with different resolutions to construct a tropospheric delay model in the China region. (b) In diverse GNSS applications, Sun et al. (2021) used the daily forecasting zenith hydrostatic delay (ZHD) and mapping function values from discrete models to determine the GPS-derived PWV. Ssenyunzi et al. (2023) used the VMF3 to estimated ZTD values for the tropical region of Africa. Yuan et al. (2019) applied the discrete model in real-time GNSS applications, evaluating its performance using forecast data from the European Center for Medium-Range Weather Forecasts (ECMWF). In addition, many GNSS data processing software use VMF1/VMF3 corrections for tropospheric delay, such as raPPPid (Glaner & Weber, 2023), Pride (Geng et al., 2019), Gamit (King & Bock, 2005), and GIPSY (Gandolfi et al., 2016). (c) For accuracy assessment, Lu et al. (2023) used the VMF3 ZWD product as a reference value to assess the accuracy of the tropospheric delay network model. Ding and Chen (2020) conducted an accuracy assessment of the empirical troposphere model GPT3 based on NGL's global tropospheric products. Yang, Meng, et al. (2021) utilized the Site-wise products as references to assess the accuracy of the proposed ZTD refinement models.

However, the research on the accuracy of the ZTD values derived from these models of the VMF data sever is still limited to our knowledge. J. Li et al. (2024) assessed the accuracy of the ZTD product from Vienna University of Technology (TU Wien, TUW) and German Research Center (GFZ), but only the Grid-wise ZTD was compared. On the other hand, Yang, Guo, Li, et al. (2021) conducted the most relevant research, which focused on the impact of differences in horizontal resolution of the models on the accuracy of the ZHD and the ZWD. However, this study utilized the ECMWF reanalysis data as the references which is the modeling data for these models of VMF data server, and only explored their performance in China. Therefore, to comprehensively explore the accuracy and availability of the ZTD values from the two types of troposphere models provided by the VMF data server is urgently needed for the GNSS community from the global scale using the GNSS-related data as references.

In this study, we conducted a comprehensive evaluation and analysis of the two types of troposphere models provided by the VMF data server based on the International GNSS Service (IGS) sites worldwide. This article is organized as follows. Section 2 describe the data collection, including VMF discrete model, VMF empirical model and the ZTD provided by IGS. The comparisons and analysis are conducted in Section 3. In Section 4, we

discuss the performance of each model using six sites with the same longitude and different latitudes. The conclusions are summarized in Section 5.

2. Materials and Methods

2.1. VMF ZTD Models

The VMF data server offers two types of troposphere delay models, that is, the discrete model and the empirical model. The discrete model consists of two sets of ZTD model, which are the Grid-wise model covering any location worldwide and Site-wise model for specific GNSS, DORIS, and VLBI stations. These discrete models are based on the direct ray tracing using the meteorological reanalysis data from ECMWF, and provides two versions including the VMF1 and VMF3 for applications (Boehm & Schuh, 2004). As an update to the VMF1, VMF3 is characterized by tuning for the specific elevation angle of 3° and providing tropospheric data with a time resolution of 6 h. Considering the advantages of VMF3 compared to VMF1, the new version is selected in the following analysis.

Specifically, the Grid-wise discrete model provides four files per day corresponding to UTC 0:00, 6:00, 12:00 and 18:00, each containing global ZHD and ZWD values with a resolution of 1°*1°. By downloading the corresponding continuous discrete file, the ZHD and ZWD values for the current epoch can be obtained by performing a linear time interpolation from the surrounding two epochs. Note that the elevation corrections were conducted in the four closest grid points to match the height of the target location. Then, the bilinear interpolation was utilized to convert the values of the four closest grid points to the target location (Yuan, 2015). The Site-wise discrete model provides the corresponding site-related daily files for some specific GNSS, Doris, and VLBI stations. Each file contains the ZHD and ZWD values at four UTC epochs, namely 0:00, 6:00, 12:00 and 18:00.

Note that there are three types of VMF discrete products providing in the data server, including VMF3-EI, VMF3-OP, VMF3-FC. The VMF3-EI utilize the previous generation of reanalysis data to generate the grid ZTD and has stopped updating. The VMF3-FC is designed as a forecast product, ideal for studies that require real-time data application. The VMF3-OP is the most widely used product in various software mentioned-above, and its accuracy assessment is the focus of this study.

The GPT3 model is the latest version of the empirical model, which was initially proposed by Landskron and Böhm (Landskron & Böhm, 2018). The GPT3 model is a comprehensive and sophisticated empirical troposphere delay model, which does not require downloading discrete files and can be used directly by inputting the time and coordinates of the sites as parameters. Note that the GPT3 model does not provide ZHD and ZWD values directly but provides various meteorological parameters, the formula of which is as follows (Lagler et al., 2013):

$$r(t) = A_0 + A_1 \cos\left(\frac{\text{doy}}{365.25}2\pi\right) + B_1 \sin\left(\frac{\text{doy}}{365.25}2\pi\right) + A_2 \cos\left(\frac{\text{doy}}{365.25}4\pi\right) + B_2 \sin\left(\frac{\text{doy}}{365.25}4\pi\right) \quad (1)$$

where $r(t)$ represents the meteorological parameters to be estimated; doy denotes the day of the year; A_0 represents its average value; and A_1 , B_1 and A_2 , B_2 are their annual and semiannual amplitudes, respectively. Then the relevant meteorological parameters were substituted into the Saastamoinen model to obtain the required ZHD (Shi, Xu, et al., 2023; Yang, Gong, et al., 2023). The corresponding formula is shown as follows:

$$\text{ZHD} = 2.2768 \times P / (1 - 0.00266 \cos 2\varphi - 0.00028h) \quad (2)$$

where φ and h denote the latitude and ellipsoidal height of the site, respectively (Yang, Sun, et al., 2023). The equation proposed by Askne and Nordius was adopted to compute ZWD (Yang, Guo, Meng, et al., 2021), which is shown as follows:

$$\text{ZWD} = 10^{-6} \times \left(k'_2 + \frac{k_3}{T_m}\right) \times \frac{R_d \times e}{g_m \times (\lambda + 1)} \quad (3)$$

where k'_2 and k_3 are the atmospheric refractive index constants with the values of 22.97 K/hPa and 375,463 K²/hPa, respectively. R_d is the specific gas constant for dry constituents, which equals 287.0464 JK⁻¹kg⁻¹ and g_m is

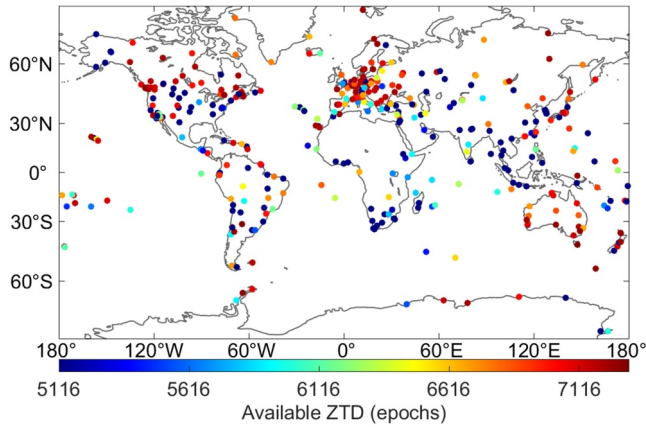


Figure 1. The distribution of global navigation satellite system sites and their available Zenith tropospheric delay data.

the average gravity that equals 9.80665 ms^{-2} . Tm and λ refer to the weighted average temperature and water vapor decrease factor, respectively (Yang et al., 2022).

Based on the understanding of the discrete and empirical models mentioned above, the subsequent comparative analysis is conducted for the site wise model, the grid wise model and the GPT3 model using data from 2016 to 2020.

2.2. IGS-ZTD Data

Since 1997, the IGS has provided the ZTD products for global monitoring stations. The IGS analysis groups including GFZ, Jet Propulsion Laboratory (JPL) and United States Naval Observatory (USNO) have generated the final IGS troposphere solution using different methods. From 1997 to 2003, independent products from multiple analysis centers were generated and combined using a weighted average method in SINEX format, with an accuracy of about 4 mm and a time resolution of 2 h (Dow et al., 2009). Since

2003, the JPL used the GIPSY software with precise point positioning (PPP) and Neil mapping function (NMF) to obtain a new ZTD product. The official accuracy of this product is 1.5–5 mm, but the actual error is slightly larger (Byun & Bar-Sever, 2009). Currently, the USNO has been responsible for producing the final IGS ZTD product using Bernese GNSS Software 5.2. This process uses PPP and the Global Mapping Function (GMF), with IGS final satellite clock/orbit and Earth orientation parameters as input. To match the site information provided by the Site-wise model, a total of 495 stations continuously updating from 2016 to 2020 were selected as references, the distribution of which is shown in Figure 1. Note that 156 station were removed in the subsequent analysis due to their missing data exceeds 30%.

2.3. Statistical Indicators

In this study, two indicators, including Bias and Root Mean Square Error (RMSE), were selected to evaluate the ZTD performance of the discrete and empirical models using MATLAB. The corresponding equations are as follows:

$$\text{Bias} = \frac{1}{N} \sum_{i=1}^N (ZTD_i - ZTD_i^R) \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (ZTD_i - ZTD_i^R)^2} \quad (5)$$

where ZTD_i and ZTD_i^R respectively represent the atmospheric parameters from four models and the reference. i and N refer to the i th value and the total number of samples.

3. Comparisons

The estimated ZTD values derived from the three models of the VMF data server and the corresponding reference ZTD values at the all sites are counted and shown as a scatter diagram in Figure 2, in which the black dashed and red solid lines represent the 1:1 straight line and the linear fit line between the estimated and reference ZTD, respectively. It is evident that the scatter plots of the ZTD values of the Grid-wise, Site-wise and GPT3 model are mostly concentrated around the 1:1 line. The correlation coefficients of the Grid-wise and Site-wise model were 0.9973 and 0.9979, respectively, which is better than that of the GPT3 model. The RMSE of the GPT3 model, Grid-wise and Site-wise model were 26.34, 8.90 and 8.00 mm, respectively, and their biases were 1.74, -0.14 and 0.63 mm. The best performance among these three models is the Site-wise model, which indicates that the estimates ZTD values of the discrete model are closer to the reference ZTD values compared to those of the empirical model on a global scale.

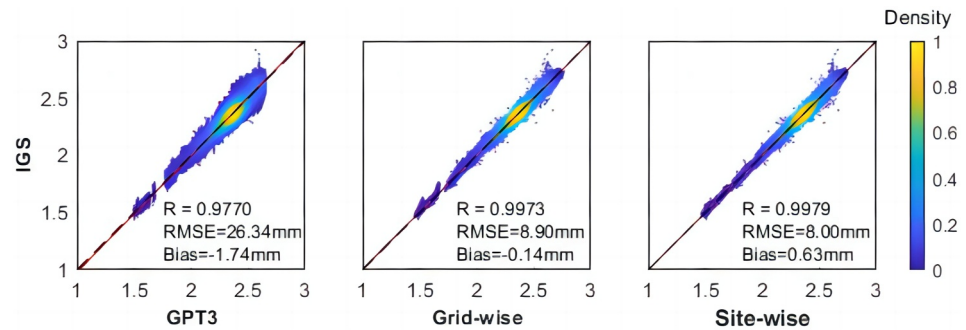


Figure 2. Scatterplot of correlation between the three models and the reference Zenith tropospheric delay.

The statistics at each site for the three models were also computed, and their average, maximum and minimum values were listed in Table 1. In terms of the average values of RMSE, Site-wise model has the smallest value, followed by the Grid-wise and GPT3 model, and their values are 11.71, 13.03, and 38.56 mm, respectively. Even though the Grid-wise model obtains the best average bias value of 0.2 mm, the Site-wise model still performs the best with the smallest bias ranging from -17.99 to 18.70 mm. For the Grid-wise and GPT3 model, the ranges are from -27.04 to 18.46 mm and from -22.82 to 24.17 mm, respectively. Comparing the RMSE values for the three models, it was found that Site-wise model has a maximum value of 23.6 mm and a minimum value of 5.05 mm. These values are superior to the Grid-wise and GPT3 model, which range from 6.02 to 31.60 mm and 18.14–68.54 mm, respectively.

To further demonstrate the differences in accuracy between the discrete and empirical models, the residual values of ZTD were calculated by subtracting the reference value from the estimates of the three models, and their histograms are shown in Figure 3. The data concentration of Grid-wise and Site-wise model in the discrete models are more pronounced compared to the empirical model, with more than 98% of the residual values distributed between -0.08 and 0.08 m. Comparing the average values of the three models, the maximum and minimum values appeared in the GPT3 and Grid-wise model with values of -2.43 and -0.15 mm, respectively. For the standard deviation, the maximum and minimum values appeared in the GPT3 and Site-wise model with values of 39.49 and 12.03 mm, respectively. These findings further indicate that the correlation of the empirical model is inferior to that of the discrete model.

Figure 4 illustrates the global geographic distribution of RMSE and bias for the three models. As shown in the left panel, the RMSE of the discrete model generally falls within the range of 10–20 mm, while the RMSE of the empirical model typically ranges from 30 to 60 mm. The empirical model exhibits a poorer performance compared to the discrete model. In addition, the accuracy of both the discrete and empirical models improves as the latitude increases, suggesting that there is a latitudinal difference in ZTD calculation, which is not reflected in the longitude. The right panel of Figure 4 demonstrates the bias of the ZTD calculations of the three models on a global scale. The bias of the empirical model exhibits a more pronounced variation with latitude, with the bias at high latitudes generally ranging from -5 to 5 mm, and the bias at low latitudes typically ranging from -20 to -10 mm. This shows that the bias of the GPT3 model decreases with increasing latitude. In contrast, the discrete model does not show a clear trend on the latitude scale, demonstrating overall greater stability. For the same latitude band, the bias of the discrete model is smaller than that of the empirical model. Moreover, the empirical model tends to exhibit cold biases in low-latitude regions, while in high-latitude areas, warm biases are more prevalent in high-latitude areas. This phenomenon is also showed in the performance of the discrete models, but note that the warm bias appeared in mid-latitude regions especially in European while the empirical model more commonly exhibits cold bias in these areas.

Table 1
The Average RMSE and Bias of the ZTD Differences for the Three Models

	RMSE (mm)			Bias(mm)		
	Max	Min	Average	Max	Min	Average
GPT3	68.54	18.14	38.56	24.17	-22.82	-2.51
Grid-wise	31.60	6.02	13.03	18.46	-27.04	-0.20
Site-wise	23.60	5.05	11.71	18.70	-17.99	0.93

To better describe the performances of the three models in different latitudes, these RMSE and bias were grouped into individual latitude bins of 10° , that is, all sites located at 80°N – 70°N were evaluated as a single unit. In Figure 5, the x-axis represents the change of latitude from north to south, and the RMSE and Bias values of each latitude band are illustrated with different colors

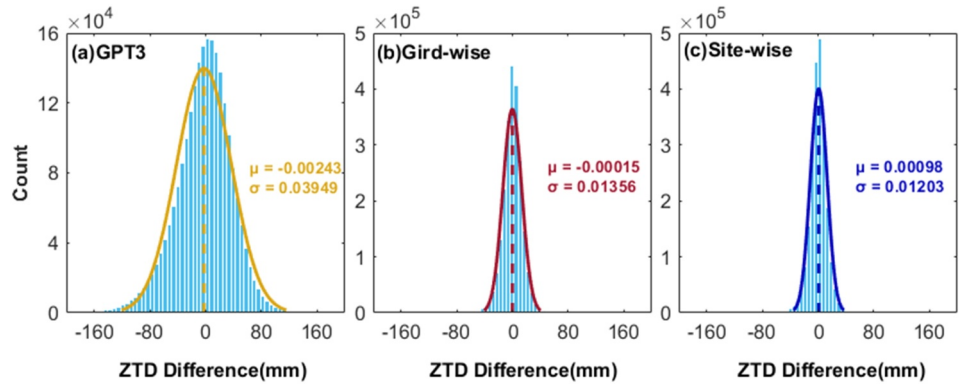


Figure 3. Residuals of the three models with reference values.

representing the three models. From the top panel, the RMSE distribution shows that the accuracy of the empirical model is lower at low latitudes than at high latitudes. The RMSE of the discrete and empirical model exhibits a tendency of decreasing with the elevation of latitude. The maximum RMSE of the empirical model is located in the region between 10°S and 20°S, with a value of 48.77 mm, and the minimum value is found between 70°S and 80°S, with a value of 23.43 mm. The RMSE of the Grid-wise model has a maximum value of 18.66 mm and a minimum value of 7.21 mm, occurring in the regions between 10°N–20°N and 70°N–80°N, respectively. The RMSE for Site-wise model has a maximum value of 16.62 mm and a minimum value of 6.59 mm, occurring in the regions between 0°S–10°S and 70°N–80°N, respectively. From the lower panel, it can be further demonstrated that the bias of the empirical model performs worse at low latitudes compared to high latitudes, while the discrete model effectively improved this phenomenon. The coldest bias of the empirical model appeared in the latitude band 0°–10°S, with a value of –10.33 mm. For the discrete model, both the Grid-wise and Site-wise model coldest biases are observed in the latitude band of 70°S to 80°S, with values of –12.61 and –7.22 mm, respectively. Moreover, the cold biases of the empirical model exceed the warm bias and are mainly distributed in the range of 60°N–60°S. The cold biases of the discrete model are primarily distributed in low latitudes, that is,

30°N–30°S, while the warm biases are mainly in the middle and high latitudes. A noteworthy phenomenon is the large bias exhibited by both models of the discrete model within the range of 60°S to 80°S. This could potentially be attributed to a limited number of stations within this latitude bin. The ZTD accuracy obtained from the three types of models exhibits variability across different latitudes. This is because low latitudes often have sufficient water vapor and relatively large ZTD values, which makes it more difficult for all the three models to characterize the ZTD changes in these latitudes, leading to poorer accuracy compared to high latitudes.

To explore the relationship between ZTD estimated by the three models and elevation, the GNSS stations were divided into three types according to their elevation, for example, <100 m (E1), 100–400 m (E2), and >400 m (E3), and the numbers of the corresponding stations were 136, 100 and 103, respectively. The RMSE and Bias values for the three models were computed in these ranges, which could be found in Figure 6. The histogram reveals that the average RMSE for the empirical model is much larger than the discrete model at any elevation, with the Grid-wise model of the discrete model being slightly larger compared to the Site-wise model. The average RMSE of the empirical model is 42.57, 38.46, and 33.37 mm for E1, E2, and E3, respectively, indicating that the average RMSE of the empirical model decreases as the elevation increases. The average bias of the empirical model also shows that it performs best in the E3 range, and it may be due to that the ZTD values are generally smaller as the elevation increases, that resulting in smaller errors. For the two types of the discrete model, the Grid-wise model has the worst average RMSE value of 14.19 mm at the elevation range of E3, and the

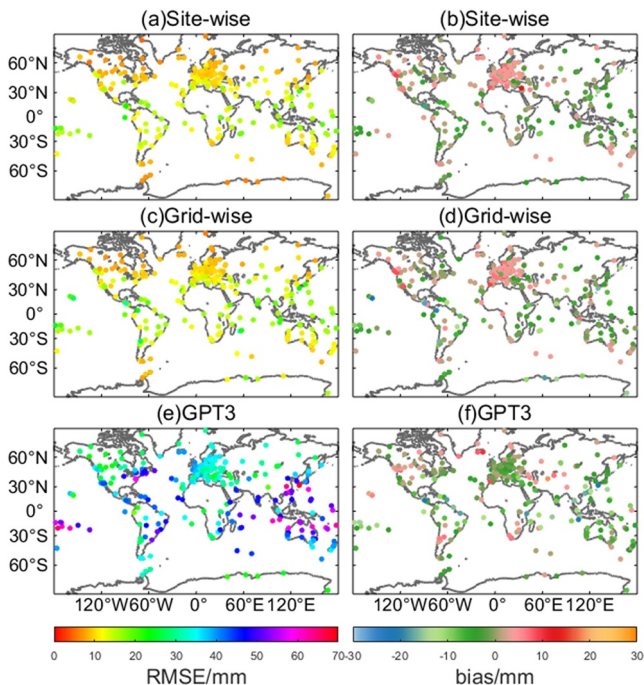


Figure 4. Geographic distribution of RMSE and bias between the three models and the reference Zenith tropospheric delay.

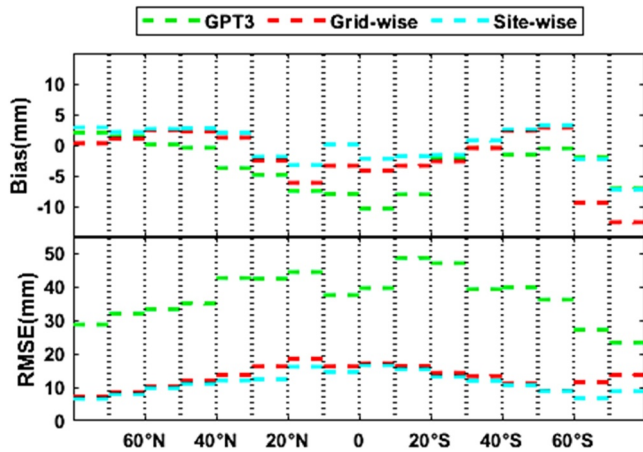


Figure 5. Distribution of average RMSE and bias over latitude for the three models.

corresponding value of the Site-wise model is relatively small in E3. This shows that the Site-wise model has better performance compared to the Grid-wise model, particularly in areas with higher elevations. In addition, the average bias for the empirical model on E1, E2, and E3 are -4.07 , -2.7 , and -0.26 mm, respectively. This indicates that the bias of the empirical model decreases with increasing elevation and consistently exhibits a cold bias. These values are $1.06/1.25/0.45$ mm and $0.22/0.71/-1.64$ mm for the Site-wise and Grid-wise model, respectively. Different from the empirical model, the discrete model does not show a clear trend in elevation change, and all biases are present warm bias. It should be noted in particular that both of the empirical model and the Site-wise model exhibit the minimum values of bias in the E3 range, while the largest value of bias appeared in the E3 range for the Grid-wise model. The value of ZTD is affected by altitude due to the atmospheric layer thickness, temperature gradients and terrain characteristics. As altitude increases, the thickness of the atmosphere temperature, and water vapor content generally decrease. These factors typically lead to lower ZTD values at higher elevations, thereby affecting the accuracy of the three models.

The differences between the ZTD estimated by the three models and the reference ZTD at all stations were grouped into different hours, namely UTC 0:00, 6:00, 12:00 and 18:00. The RMSE of different months were computed epochs, and the monthly variation of ZTD-RMSE at these four epochs were shown in the form of linear heat map in Figure 7. The RMSE derived from the three models exhibit the same seasonal differences, with the worst performance in June–August and the best performance in December–February. Specifically, these three models exhibit the largest RMSE around July and all show a downward trend from July to December and an upward trend from January to July. For the discrete model, the ZTD-RMSE exhibited significant differences when calculated at different epoch on the same day. The difference in the ZTD-RMSE of the discrete model at the four UTC epochs under the same month is pronounced, especially in June–August and March–May, which has the worst performance at 00:00 and the best performance at 06:00. In contrast to the discrete model, the empirical model shows no obvious different performances at different UTC epochs. The difference in data accuracy between different UTC epochs can be attributed to the diurnal ZTD variation, which means that meteorological parameters such as temperature, atmospheric pressure, and humidity that affect the magnitude of ZTD value are changing at different epochs. The large ZTD values results in relatively poor accuracy of the discrete models at the corresponding epochs. For the GPT3 model, it is an empirical model that only contains the characteristics of ZTD annual and semiannual cycles. Thus, the performance of the GPT3 model is similar at different epochs.

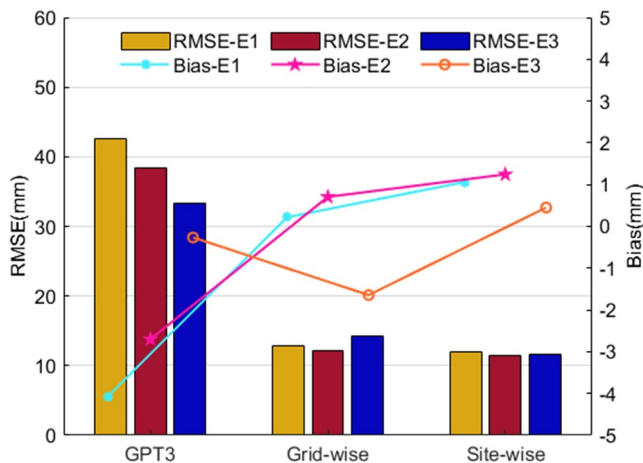


Figure 6. Average RMSE and Bias of Zenith tropospheric delay at different elevations (E1, E2, E3 indicate elevation ranges, <100 m, $100\text{--}400$ m, >400 m).

To further illustrate the performance of the three models in different seasons, a division was made for every 3 months to obtain S1 (December to February), S2 (March to May), S3 (June to August), and S4 (September to November), and the RMSE and bias between the three models and the reference data were counted and shown in Figure 8, in which the upper and lower panel represent the box plot of bias and the histogram of the average RMSE, respectively. The median values of bias for the empirical model are -1.71 , -3.16 , -2.18 and -2.28 mm for the four seasons, respectively, all of which are more unfavorable than the corresponding values for the discrete model. These results indicate a consistent cold bias in the empirical model throughout all four quarters, with the worst bias occurring in S2 and the best in S1. For the two types of the discrete model, the Site-wise and Grid-wise models of the discrete model exhibit biases of $1.04/0.55/0.82/1.46$ mm and $-0.38/-0.66/0.20/0.23$ mm, respectively. The Site-wise model demonstrates a warm bias across all quarters, while the Grid-wise model does not display a discernible tendency toward either warm or cold bias. The lower panel effectively captures the seasonal variation in accuracy for both the discrete and empirical models, aligning with the trends depicted in the linear heat map in Figure 7. All of these three models exhibit the largest average RMSE in the S3 quarter

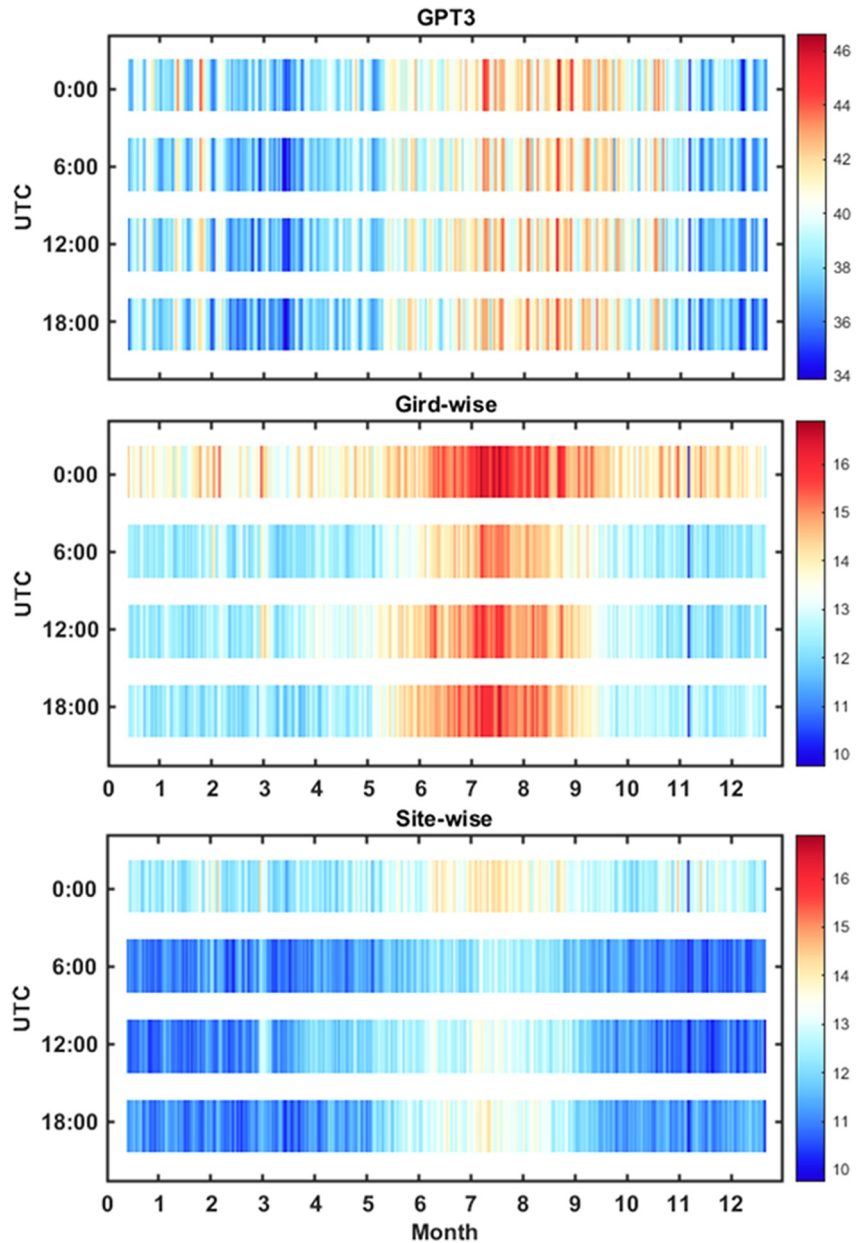


Figure 7. Linear heat map of RMSE for three model months and times.

with the values of 12.97, 14.82 and 40.86 mm, respectively. The best accuracies for the Grid-wise and Site-wise model are observed in the S1 quarter with values of 12.77 and 11.45 mm, respectively. In contrast, the GPT3 model exhibits its highest accuracy in the S2 quarter, with a value of 37.92 mm.

4. Discussion

To gain further insights into the patterns and interrelationships between the discrete and empirical models, the time series of ZTD estimated by the three models and ZTD reference values as well as their difference were plotted in Figure 9, in which six GNSS stations situated at different latitudes with similar longitudes (69°W) for the period of 2016–2020 were selected. Specifically, the six stations locate in the range of 60°N–90°N (THU2), 30°N–60°N (UNBJ), 0°–30°N (RDSD), 60°S–90°S (ROTH), 30°S–60°S (MGUE), and 0°–30°S (AREQ), respectively. In Figure 9, the dots of different colors correspond to different models, the blank part indicates the missing ZTD reference value, and the black dashed line represents the average value of the ZTD reference value.

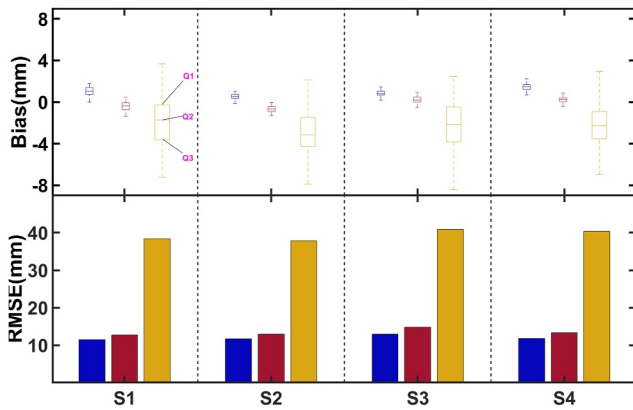


Figure 8. Seasonal differences in average RMSE and bias for the three models.

The results illustrate that both ZTD estimates and reference values exhibit regular variations. The ZTD estimates from the empirical model exhibit a consistently periodic pattern over time, attributed to their derivation from empirical meteorological parameters grounded in annual meteorological conditions. Bias analysis reveals that the GPT3 model exhibits a more scattered bias in comparison to the Grid-wise and Site-wise models. The degree of bias aggregation increases with the latitude, which is evidenced by the fact that the sites located in high-latitude (a and d) show bias concentrated around 0, while the sites located in low-latitude (c and f) display more dispersed bias around 20. In addition, the bias scatter of Grid-wise and Site-wise models exhibits minimal overlap at high latitudes, while the sites located in mid-latitude and low-latitude demonstrate considerable overlap in bias scatter points. The difference is observed might be attributable to the unique topographical features in high latitudes, such as glaciers and permafrost. They can result in spatial differences in the environmental and atmospheric conditions of these areas. Additionally, extreme seasonal and weather changes occur in high-latitude areas, which also affect atmospheric conditions. The Grid-wise

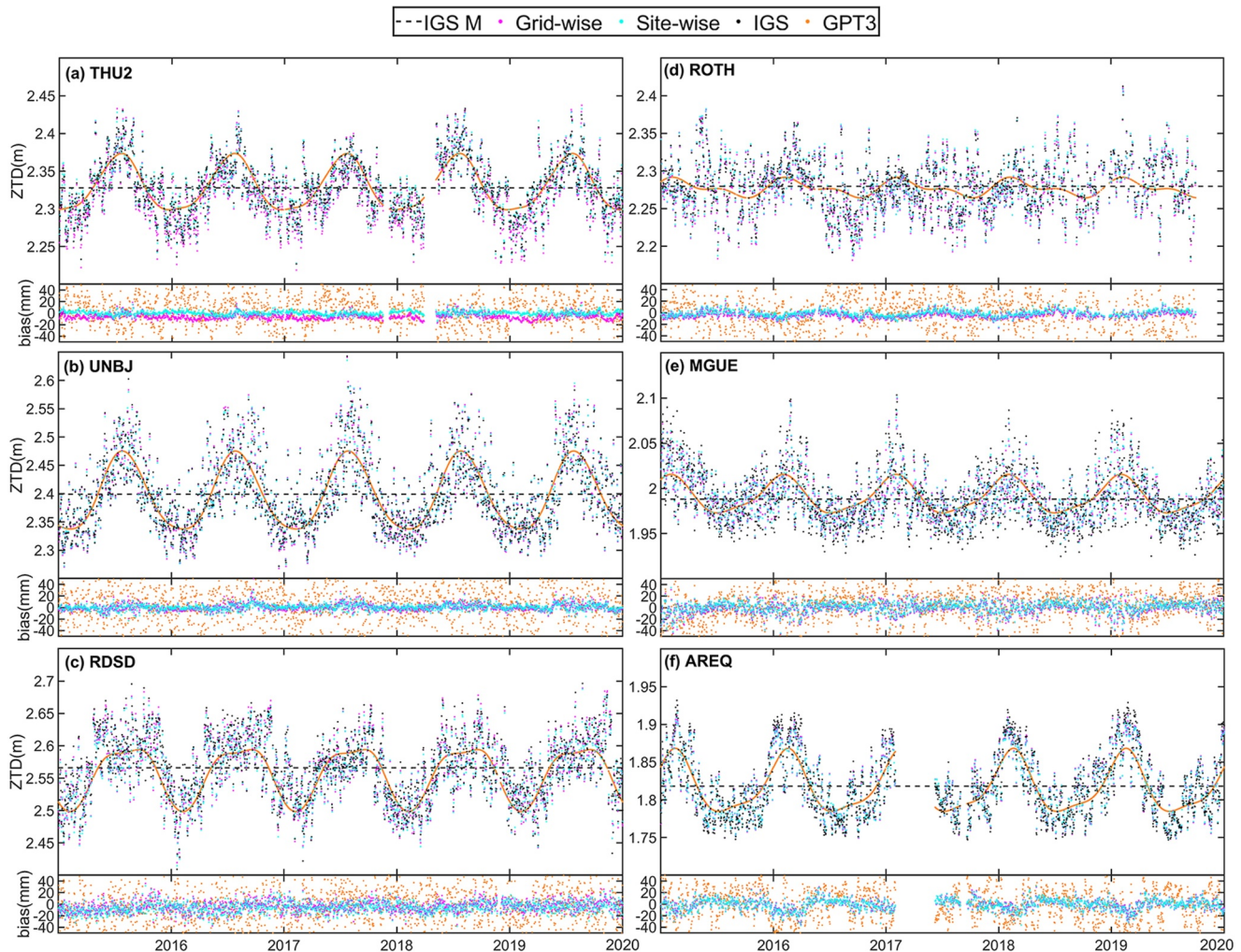


Figure 9. Time-series analysis of three models and Zenith tropospheric delay reference values for six different latitudinal zones (from high northern latitudes to high southern latitudes).

model interpolates data spatially and has good spatial generalization capabilities, but it cannot capture spatial regional variations and seasonal impacts as effectively as the Site-wise model, then leading to errors.

Figure 9 also illustrates some other phenomena in addition to those mentioned above. Specifically, the Southern Hemisphere demonstrates lower volatility in both ZTD estimates and reference values compared to the Northern Hemisphere, with the smallest volatility occurring at the site located in 60°S–90°S (ROTH). In addition, the sites located in the northern and southern hemispheres have different trends, that is, the average value of the ZTD reference value at the sites located in northern hemisphere show an increasing trend from high to low latitudes, while the sites located in southern hemisphere show the opposite trend, with ZTD averages decreasing at low latitudes. Note that the above phenomenon is all based on the results of the selected six stations, and whether it is a global common phenomenon still needs to be better explored if more abundant stations could provide corresponding data in the future. The focus of this article is to compare the accuracy of the three types of ZTD models, and the detailed characteristics of ZTD itself with latitude changes need to be further explored in follow-up research.

5. Conclusions

This article focuses on the analysis of zenith troposphere delay based on discrete and empirical models obtained from VMF Data Server. To comprehensively compare the performance of these models, we employed the ZTD products provided by IGS as reference values. Subsequently, we calculated the RMSE and bias for each of the three models across 495 IGS sites globally, spanning the temporal domain from 2016 to 2020. The numerical results illustrated that Site-wise and Grid-wise models exhibit similar correlations and residual distributions and both exhibit better accuracy performance compared to the GPT3 model. From perspective of spatial distribution, the RMSE of ZTD for all three models was notably larger in low-latitude regions, with the values of the Site-wise and Grid-wise model predominating above 15 mm and the values of the GPT3 model surpassing 35 mm. Conversely, smaller values were observed in high-latitude regions, with the Site-wise and Grid-wise model predominantly below 10 mm and the GPT3 model below 30 mm. Site-wise exhibits superior global performance compared to the other two models, boasting an average RMSE of 11.71 mm. This represented an improvement of 1.32 mm over the Grid-wise model and a substantial 26.85 mm over the GPT3 model. Furthermore, the data revealed a positive correlation between increased latitude and enhanced accuracy across all three models. Regarding spatial elevations, the GPT3 model displayed a discernible trend of diminishing RMSE and bias values with increasing elevation, a phenomenon not observed in the discrete models.

Temporal analysis unveiled distinct seasonal variations in ZTD, with larger RMSE values in summer and smaller values in spring across all three models. Site-wise consistently outperformed the other models in all four seasons, with average RMSE values of 11.45, 11.69, 12.97 and 11.80 mm, respectively. Additionally, the Site-wise and Grid-wise model exhibited variable accuracy at different UTC intervals, achieving optimal performance at UTC 06:00 and reduced accuracy at UTC 00:00. This phenomenon was not mirrored in the GPT3 model, which maintained consistent performance across varying UTC times.

To understand the patterns and relationships between the discrete models and the empirical model, the time series of ZTD estimates values of the three models, ZTD reference values and their differences for sites at different latitudes were further investigated in temporal and spatial direction. Both ZTD estimated values and ZTD reference values exhibit periodic trends over time. The ZTD calculated value of the GPT3 model was based on empirical meteorological parameters and exhibits exactly the same value in each cycle. In addition, ZTD estimates and reference values are less volatile in the Southern Hemisphere than in the Northern Hemisphere, with smaller differences between peaks and troughs.

Based on the research presented in this study, the Site-wise model is undoubtedly the best choice when the research or applications are conducted at IGS stations. For the sites not covered by the Site-wise model, the Grid-wise model is the preferred choice. When it is not possible to obtain the data files for discrete models or when the accuracy requirement of ZTD prior information is not high, the GPT3 model is recommended.

Data Availability Statement

The ZTD estimated and reference in this study are available at (Yao, 2024) <https://doi.org/10.5281/zenodo.11093896>. The original data of the VMF ZTD models can be found at <https://vmf.geo.tuwien.ac.at/>. The more data of IGS-ZTD can be accessed after registration from <https://cddis.nasa.gov/archive/gnss/products/troposphere/zpd/>.

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